

ROYAL SACHAN

Final-Year B.Tech, Computer Science – IIIT Manipur | Bengaluru, India

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Summary

Final-year CS student with backend internship experience (Node.js, PostgreSQL) who has built production RAG systems (hybrid search, cross-encoder reranking). Targeting software engineering roles focused on scalable backend systems.

Education

Indian Institute of Information Technology, Manipur

B.Tech in Computer Science and Engineering

Sep 2023 – Apr 2027

Manipur, India

Professional Experience

Backend Developer Intern

Atal Innovation Centre, Delhi

May 2025 – Jul 2025

Node.js | Express.js | PostgreSQL | JWT

- Built **JWT authentication and role-based access control** in Node.js, securing **20+ REST API endpoints** across 3 user roles (admin, manager, cashier) for a Point of Sale application
- Reduced slow endpoint response times by **30%** through PostgreSQL query optimization and composite indexing; added rate limiting, CORS headers, and brute-force protection middleware
- Designed and documented **RESTful API contracts** for the POS billing module, enabling parallel frontend development and reducing integration bugs during handoff

Projects

PaperTrail – Multi-Article RAG Research Assistant (Python | FastAPI | Weaviate | Groq | React)

 | Live

- Developed a RAG web app (FastAPI + React) that scrapes news URLs, chunks into **~300-token** segments, embeds with BAAI/BGE-small-en-v1.5, and indexes in Weaviate for source-attributed answers
- Engineered a hybrid retrieval pipeline with tuned BM25 and HNSW dense search fused via RRF, applying cross-encoder reranking (ms-marco-MiniLM) to return top-6 passages
- Implemented intent-based query routing between Groq models (Llama-3.3-70B vs. DeepSeek-R1) with SSE streaming and multi-turn dialogue over a 10-message history window
- Evaluated retrieval quality using a custom RAGAS-style LLM-as-judge framework across Faithfulness, Answer Relevancy, and Context Precision, achieving a RAG score of **0.905**

Sangam–Event Companion Platform (Java | Spring Boot | PostgreSQL | Redis | React | Docker)

 | Live

- Architected a full-stack event-discovery platform in Spring Boot with role-based group join approval, supporting concurrent request handling via async processing
- Implemented location-based event search using **PostGIS** geospatial indexing, returning distance-ranked results with **sub-100ms** query response on 10,000+ indexed locations
- Designed an automated waitlist system using a Redis-backed queue that enqueued users at capacity and triggered slot-open notifications, eliminating manual approval overhead
- Secured the application with **JWT refresh-token rotation** and Redis rate limiting; deployed via **Docker** on Railway with automated **CI/CD** via GitHub Actions

EcoRoute–Intelligent Delivery Optimizer (Python | FastAPI | MongoDB | React)

 | Live

- Engineered a route optimization engine using **K-Means++ clustering** to partition delivery locations and a **Nearest Neighbor TSP heuristic** to sequence stops, reducing average route length by **~35%** over naive ordering across 500-stop test datasets
- Modeled delivery locations as **GeoJSON Points** in **MongoDB** with **2dsphere** indexes, enabling sub-millisecond radius queries via **\$nearSphere** without a PostGIS extension
- Built a **FastAPI** backend with **Motor** async driver and **JWT authentication**, isolating the optimization computation layer from API controllers for stateless, independently scalable processing

Technical Skills

Languages: Java, Python, JavaScript, SQL

Backend & APIs: Spring Boot, Node.js, Express.js, FastAPI, REST APIs, JWT, Docker, JPA/Hibernate, Maven, Git

Databases & Infrastructure: PostgreSQL, Redis, MongoDB, Weaviate, GitHub Actions, Vector Search, Vercel

AI/ML: RAG, Hybrid Retrieval (BM25 + HNSW + Reranking), BGE, Sentence Transformers, Hugging Face, LLM APIs (Groq)

Achievements

- Secured **AIR 2** in **EROH 2025** by Naukri.com among IIT/IIIT/NIT participants; awarded merit scholarship, featured in *Times of India*
- Ranked **5,204** globally in **Meta Hacker Cup 2025**, clearing Round 1 among hundreds of thousands of participants
- Selected among **top 3 proposals** at IIIT Manipur for **5G Innovation Hackathon 2025** by DoT, Government of India
- Solved **350+ DSA problems** across [LeetCode](#) and [CodeChef](#)